University Library Management System (ULMS) Project Documentation

1. Project Overview & Scope

The University Library Management System (ULMS) is an integrated web-based platform designed to automate and streamline core operations of an academic library. The system manages books, digital resources, user profiles (students, faculty, administrative staff), book reservations, issue/return transactions, and automated library fine calculations.

Key Modules:

- **User Management & Authentication:** Role-Based Access Control (RBAC) for Admins, Librarians, Faculty, and Students.
- **Catalog & Inventory Management:** Add, update, search, and categorize physical books, journals, and digital media.
- **Circulation & Transaction Management:** Book check-out, check-in, extension requests, and automated fine calculations.
- **Online Public Access Catalog (OPAC):** Advanced search engine with filters (Author, ISBN, Title, Availability).

2. Objectives, Assumptions & Constraints

Objectives

- **Automate Manual Processes:** Eliminate paper logs and reduce librarian administrative overhead by 60%.
- **Improve Accessibility:** Provide 24/7 web-based access to the catalog and reservation system.
- **Accuracy & Accounting:** Eliminate human error in tracking due dates, extensions, and fine calculations.

Assumptions

- All users possess valid institutional email credentials for account verification.
- The system will be hosted on a cloud or university-managed web server.

Constraints

- **Timeline:** Project development must be completed within an 8-week academic schedule.

- **Budget:** Open-source tech stack ($0 software licensing cost).

3. Software Requirements Specification (SRS)

Functional Requirements

- **FR-01 (Authentication):** The system shall authenticate users via institutional email and assign roles (Student, Faculty, Librarian, Admin).
- **FR-02 (Book Search):** The OPAC search engine shall allow users to filter books by Title, Author, ISBN, and Category.
- **FR-03 (Circulation):** The system shall allow librarians to scan book barcodes and member IDs to record check-outs with a default 14-day loan period.
- **FR-04 (Fine Calculation):** The system shall automatically calculate overdue fines at a rate of $0.50 per day past the due date.
- **FR-05 (Reservations):** Students shall be able to place a hold/reservation on checked-out books and receive email notifications upon return.

Non-Functional Requirements

- **NFR-01 (Performance):** Search query results must render in less than 1.5 seconds under standard load.
- **NFR-02 (Security):** Passwords must be hashed using bcrypt, and SQL queries must use parameterized statements to prevent SQL injection.
- **NFR-03 (Availability):** The web platform shall maintain 99.9% uptime during academic terms.

4. Project Management & Administration Reports

Work Breakdown Structure (WBS)

□1. University Library Management System
  1.1 Initiation & Planning
    1.1.1 Project Scope & Charter Definition
    1.1.2 Repository Setup & Git Workflow
  1.2 Requirements Engineering
    1.2.1 Stakeholder Analysis & Interviews
    1.2.2 SRS Document Compilation
  1.3 System Design & Modeling
    1.3.1 Relational Database Schema & ERD

Task / Deliverable	Project Lead	Frontend Dev	QA Engineer
Project Scope & Charter	A / R	C	C
SRS Document	A	C	R
Database Schema & ERD	A	I	C
UI Wireframe Prototypes	A	R	I
Backend & Frontend Coding	A	R	I
Testing & Bug Resolution	A	C	R

1.3.2 UML Use Case & Sequence Diagrams
1.3.3 UI/UX Wireframe Prototyping
1.4 Implementation (Coding)
1.4.1 Authentication & RBAC Module
1.4.2 Book Catalog & Search Module
1.4.3 Circulation & Fine Calculator Module
1.5 Testing & Quality Assurance
1.5.1 Unit & Integration Testing
1.5.2 User Acceptance Testing (UAT)
1.6 Deployment & Closure
1.6.1 Cloud Environment Deployment
1.6.2 Final Project Report Submission

□Responsibility Assignment Matrix (RACI Chart)

Project Schedule & Gantt Chart (8 Weeks)

□+----------+----+----+----+----+----+----+----+----+
| Task Description | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 |
+----------+----+----+----+----+----+----+----+----+
Initiation & SRS Gathering	████	████						
Database & UML Design			████					
UI Wireframing & Prototyping			████	████				
Core Module Development				████	████	████		



Risk Assessment & Mitigation Plan

Risk ID	Risk Description	Likelihood	Impact
R1	Scope creep during feature development	Medium	High
R2	Concurrent book reservation race conditions	Medium	High
R3	Task delays or bottlenecking from team members	Medium	Medium

5. Repository Setup & Git Execution Guide

Required Folder Hierarchy

```
university-library-management-system/
├── README.md
├── requirements/
├── diagrams/
├── ui-prototype/
└── reports/
```